

Advanced Neuro-imaging for TBI *Implications in 2016.*

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24th November 2016
Health Claims Forum

“Diagnosis and Prognosis - Hidden in Plain Sight “

Mild Traumatic Brain injury

**Role of Neuroimaging in 2016 to Identify :
Structural, functional and microscopic evidence of:**

**Traumatic brain injury
&
Risk of Progressive Neurodegenerative Brain Injury**

“Diagnosis and Prognosis - Hidden in Plain Sight “

Brain Injury – advanced neuroimaging

current and future possibilities

Mild Moderate Severe TBI

TBI “silent epidemic”

10% fatal

5-10% serious permanent deficits

20-40% moderate disability

>40% “minimal brain trauma” subtle deficits.



If only we could see & measure what is happening in the brain

MESSAGE FROM THE EDITOR

Traumatic Brain Injury:
A Glimpse of Order Among the Chaos?

CASE 1

An 18-year old high school student suffered multiple traumatic injuries after a severe automobile accident. He was admitted to the intensive care unit in coma, and brain imaging revealed soft tissue injury, asymmetric, bifrontal cerebral contusions, as well as moderate white matter changes consistent with diffuse axonal injury. Three weeks later he was minimally responsive and a senior neurointensivist stated his prognosis for a meaningful neurological recovery was essentially nil. Two years later, he walked across the stage to receive his high school diploma after completing all his required courses.

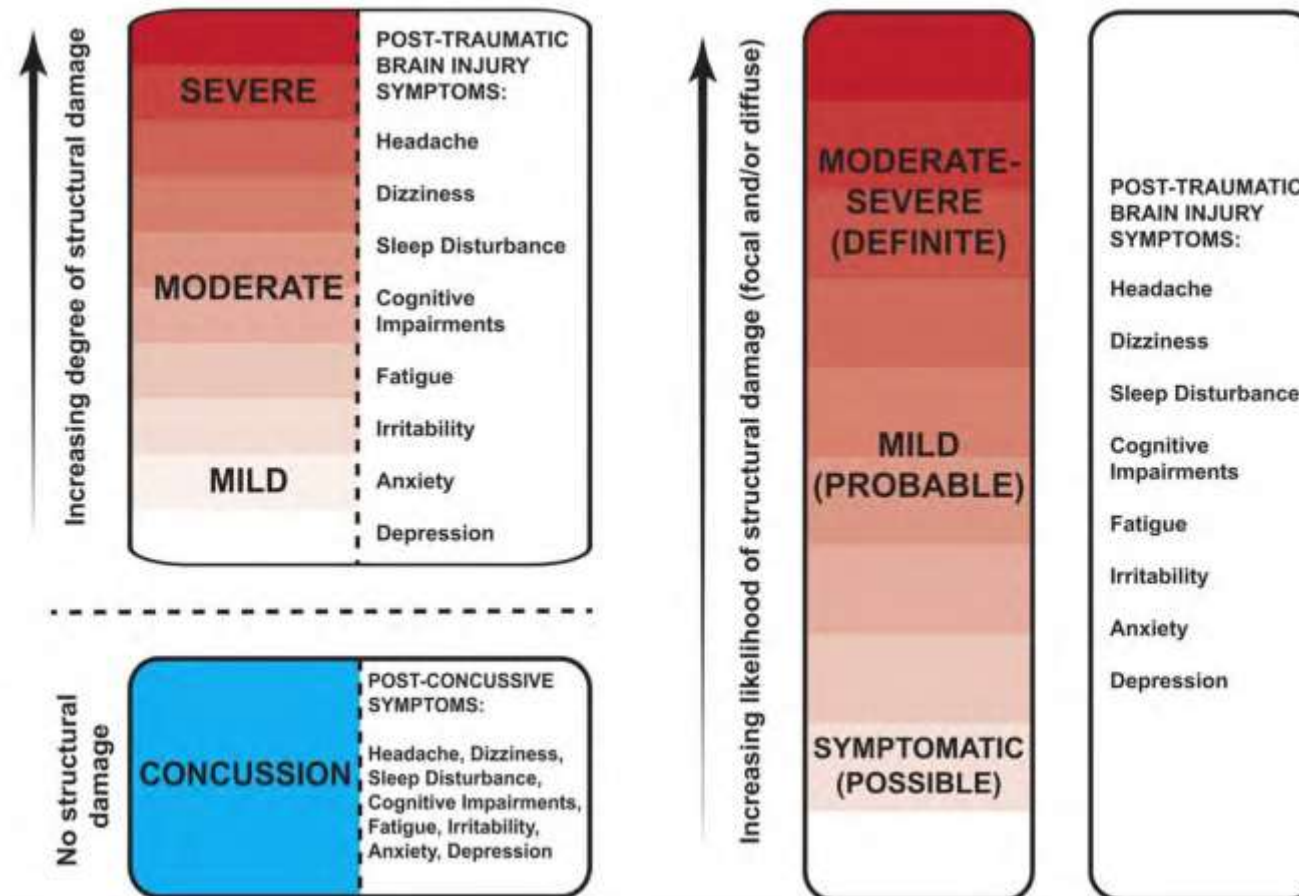
Case 1

A 35 year-old freelance writer struck his head after falling off the back of a pickup truck while working on an Alaskan pipeline project during the summer. He was unconscious for three to four minutes and taken to a local emergency department where his neurological exam was grossly normal and a brain CT scan showed no evidence of injury. Eighteen months later, at one of his regular quarterly outpatient neurology clinic appointments, he said he was finally beginning to recover his normal level of energy, alertness, and clarity of thinking, and he was encouraged by the fact that he was beginning to write again at the same level of proficiency as prior to the accident.

Concussion is confusing us all

David J Sharp, Peter O Jenkins

For neuroradiologists concussion doesn't make any sense !



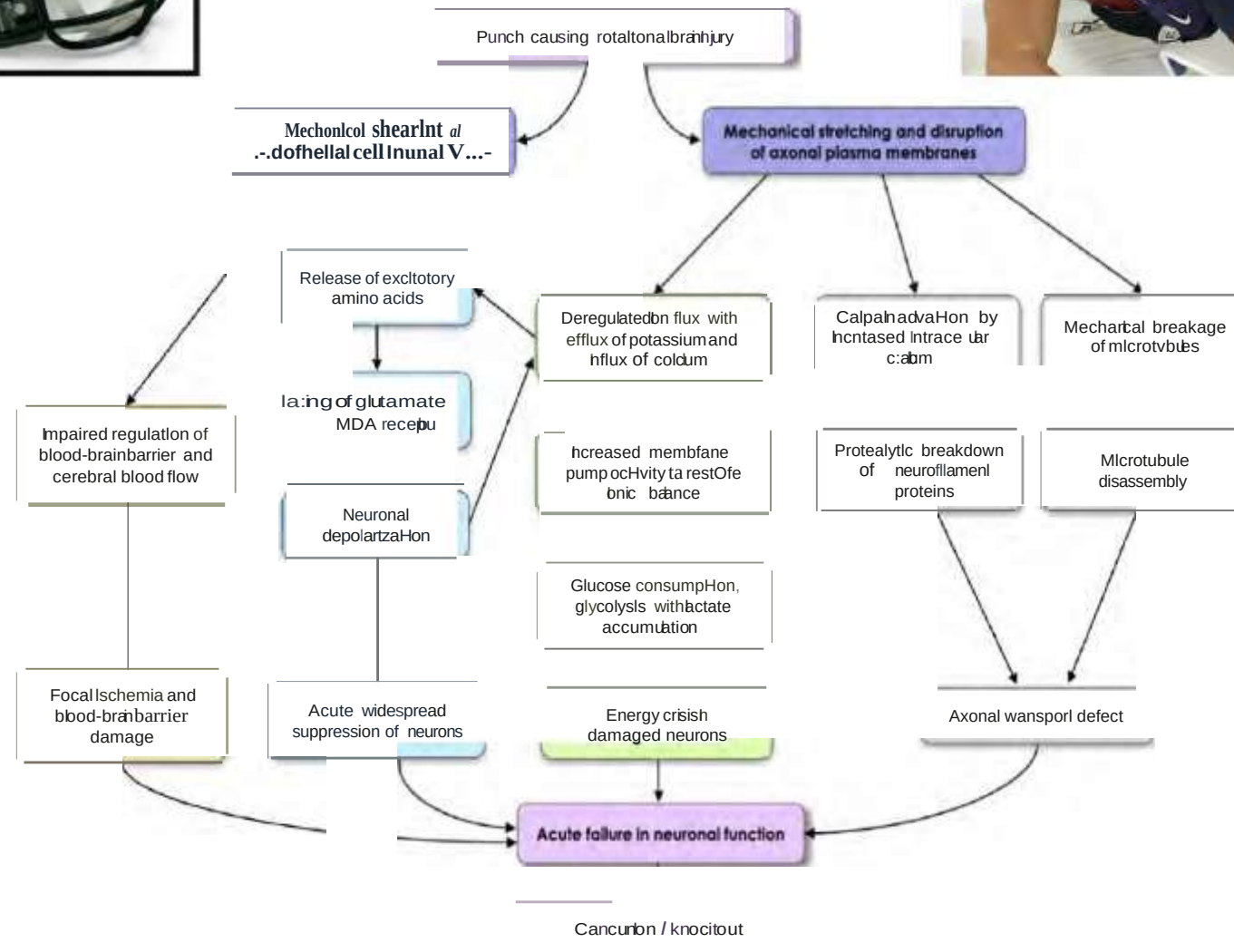
When and how to use imaging – clinically valid for M/L purposes vs research ?

- CT acute imaging, immediate medical action required
- MRI analysis of acute injury not demonstrated on CT & evaluation of neurological, cognitive and behavioural sequelae
- MRI Diffusion tensor imaging (DTI) white matter tractography
- MRI Diffusion kurtosis imaging (DKI) white matter injury and predictor of cognitive outcome
- fMRI resting state brain networks
- PET amyloid and tau ligand imaging to detect biomarkers & Chronic Traumatic Encephalopathy (CTE)



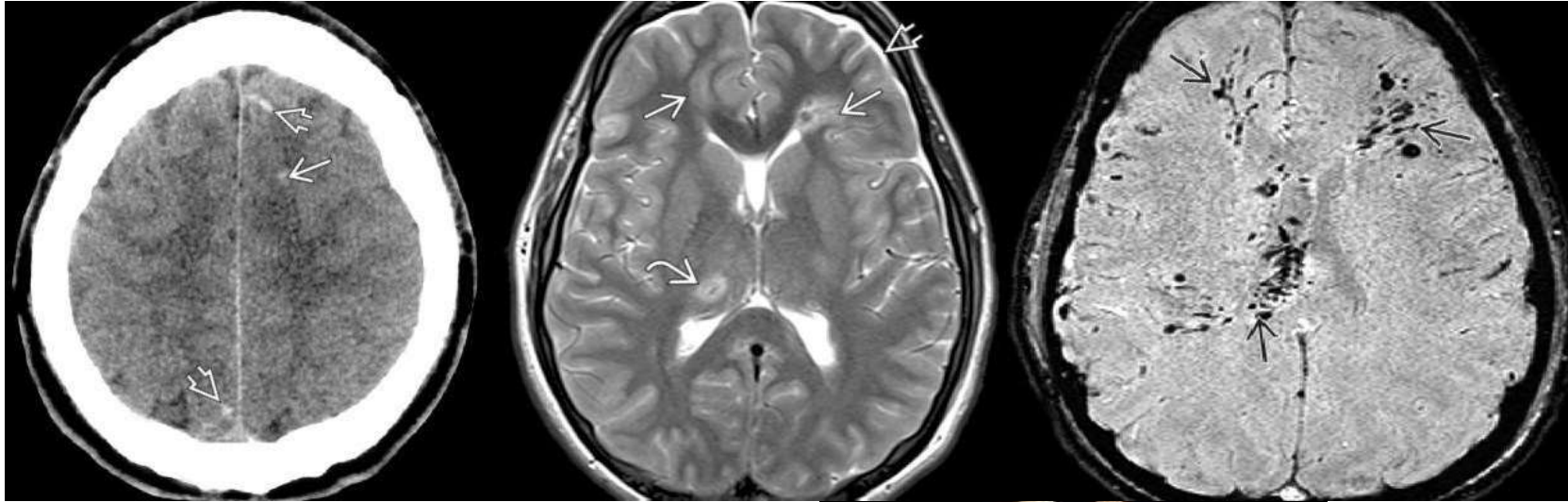


How much of this can we see ?

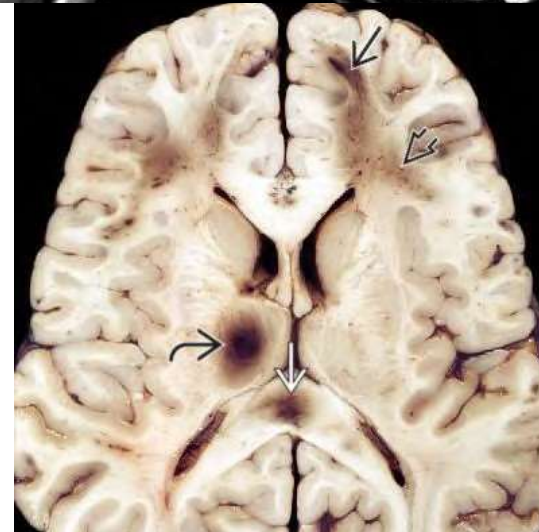


Head Injury - How to image ?

High speed RTA. GCS 8 CT did not explain symptoms.

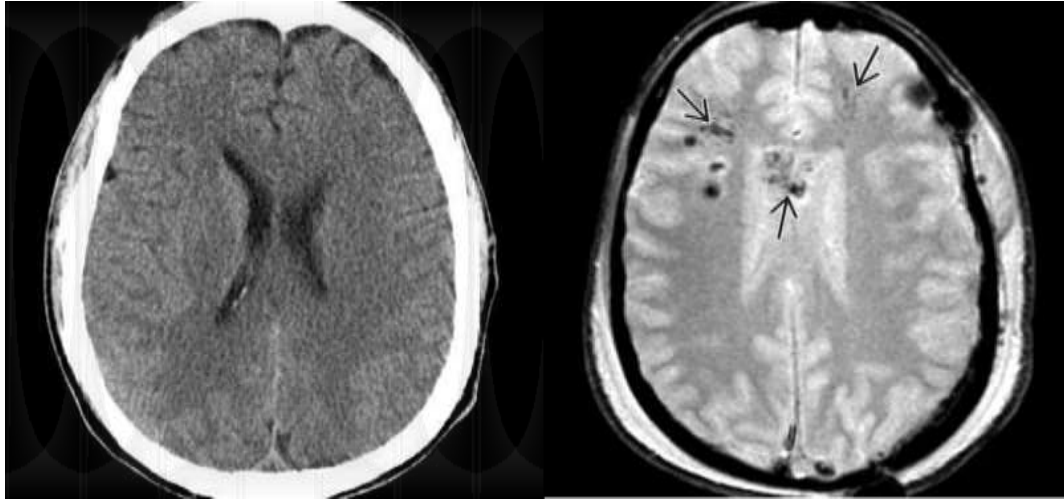


“Hidden in plain sight”

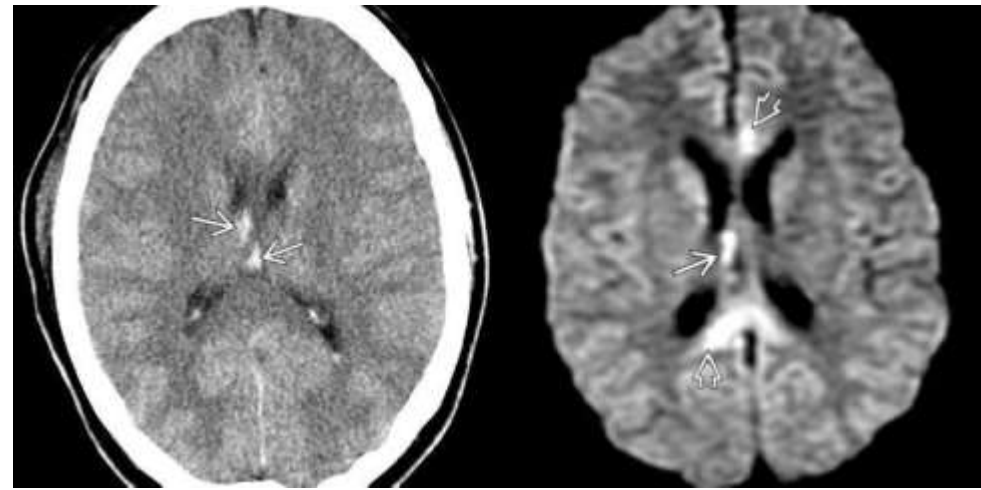


Head Injury - How to image ?

2 High speed RTAs GCS 8 CT did not explain symptoms.



“Hidden in plain sight”



When symptoms and conventional imaging don't match ?

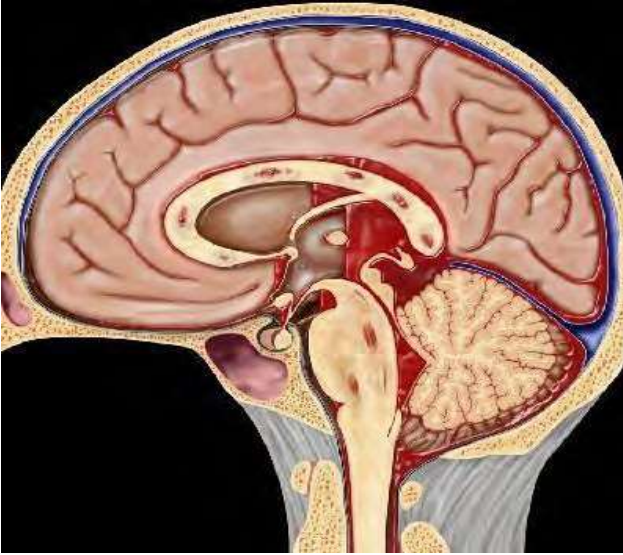
A minefield – how can diagnostic imaging help us: MRI in 2016

- **T2*GE and SWI Susceptibility Weighted Imaging**
*3x sensitivity at 3Tesla (T) for microhaemorrhages.
Long after the time of head injury, accepted biomarker for DAI.
Not visible on CT and often early MRI
Extent only appreciated on high field **3T vs 1.5T***
- **DWI Diffusion Weighted Imaging**
*Detects the damage acute phase of traumatic brain injury
Changes in the rate of microscopic water motion cell death
Associated with DAI and other mechanisms of injury.*
- **DTI Diffusion tensor imaging**
*Information on the structural integrity of white matter
Evaluation as a tool in the acute phase for predicting longer term outcome*

SWI and DWI imaging is being used increasingly for clinical assessment of **subtle brain injury and should now be accessed routinely to assess TBI.**

Primary effects of Head Injury

Diffuse axonal injury DAI



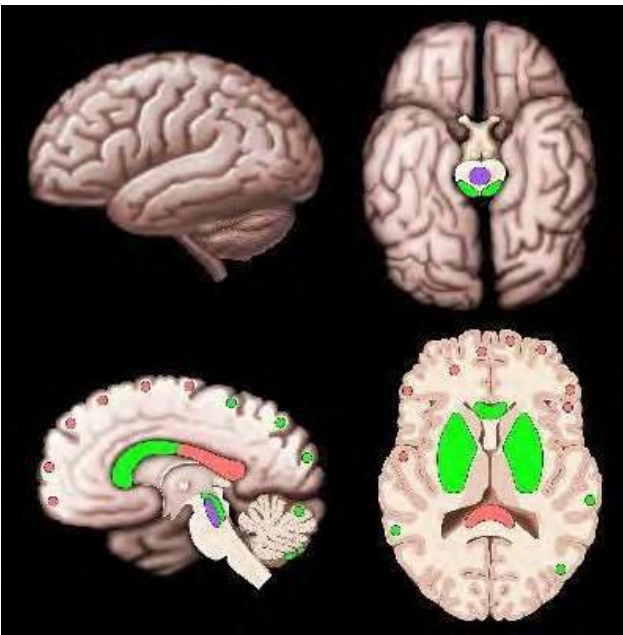
Traumatic shearing (rotational) forces

Disruption of axons = white matter

Different density tissues glide over one another
Stretching axons that traverse junctions between
different density between grey and white matter

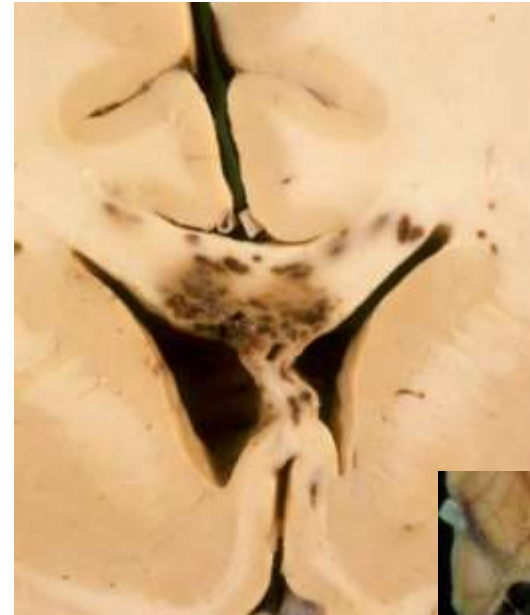
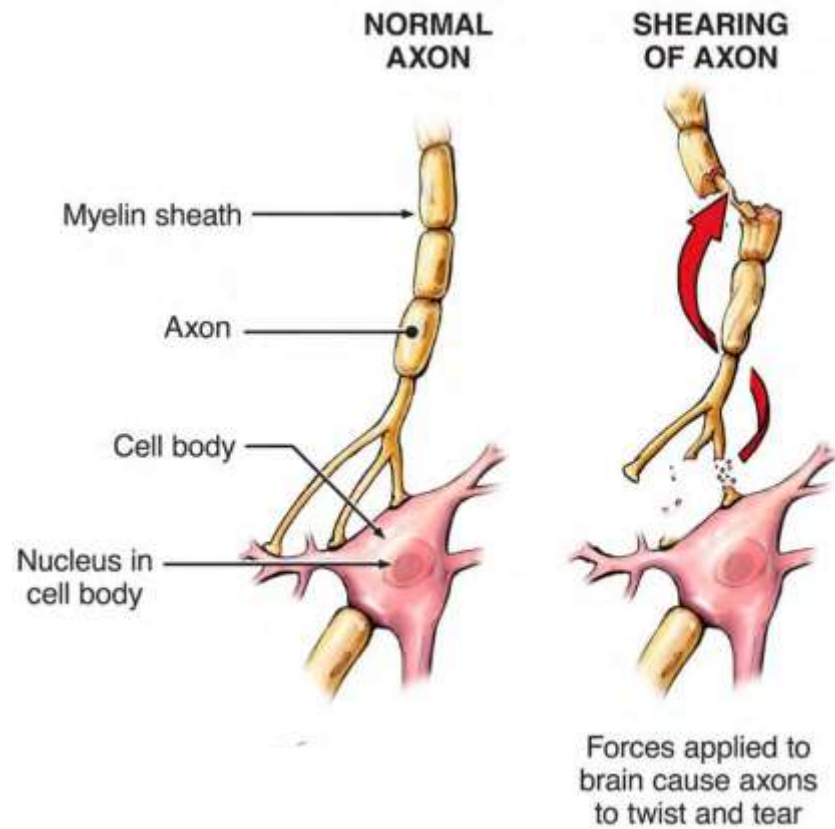
"Hidden in plain sight"

*You need to know what you are looking for
and how to find it :*



Diffuse Axonal Injury DAI

Susceptibility weighted MR imaging



Importance of detecting micro-haemorrhage in TBI

The hallmark for DAI

Comparison of TBI patients with and without micro-haemorrhages:

- *Presence of micro-haemorrhages, correlates with more severe disruption of white matter tracts, in large parts of the white matter.*
- *The relationship between white matter abnormalities and cognitive function demonstrated in two key domains i.e. memory and executive function.*

*DAI is responsible for the **majority of global cognitive defects seen after TBI**, particularly with regards to difficulties with memory and information processing.*

With respect to traumatic DAI classification:

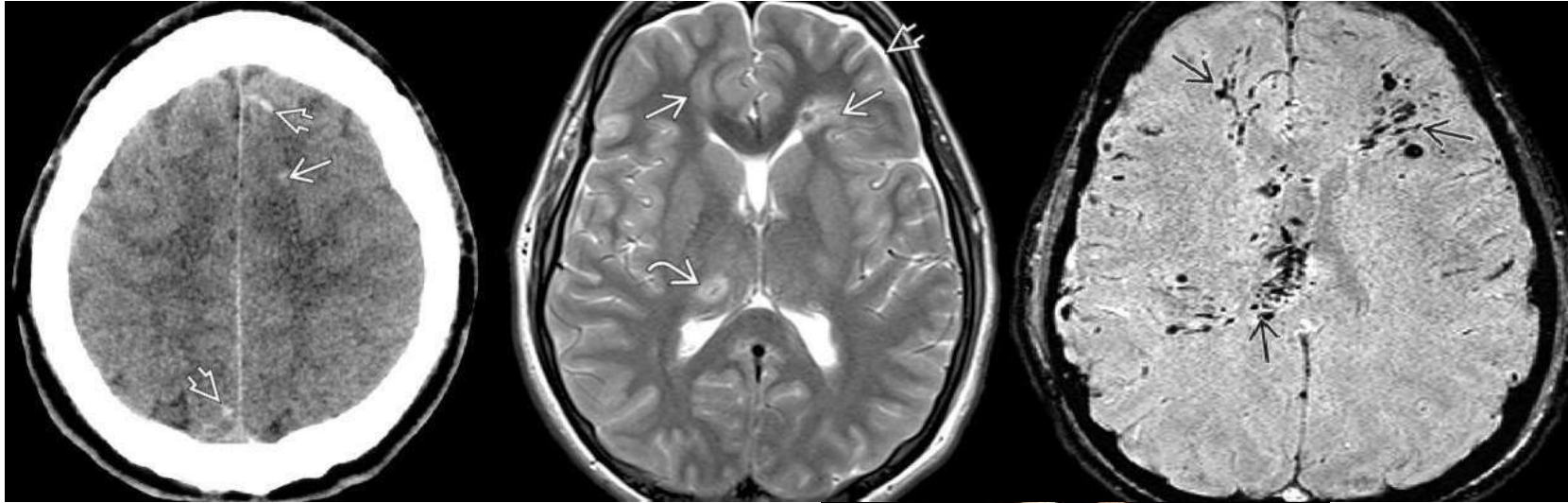
- *Mild Grade 1* *grey white matter junction of the cerebral white matter*
- *Moderate Grade 2* *grey/white matter & corpus callosum*
- *Severe Grade 3* *grey/white matter & corpus callosum & mid brain*

Head Injury - How to image ?

CT

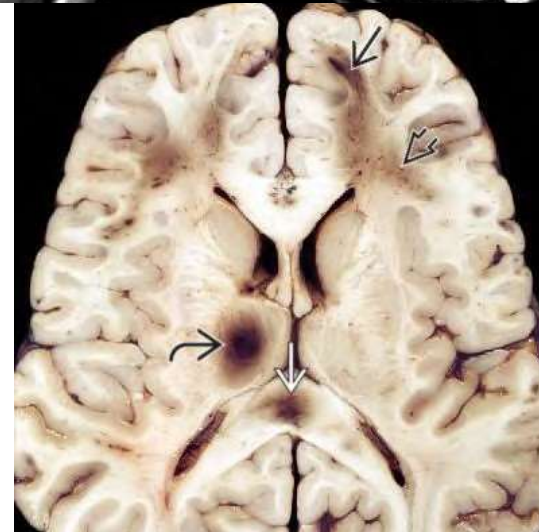
T2 MRI

SWI MRI



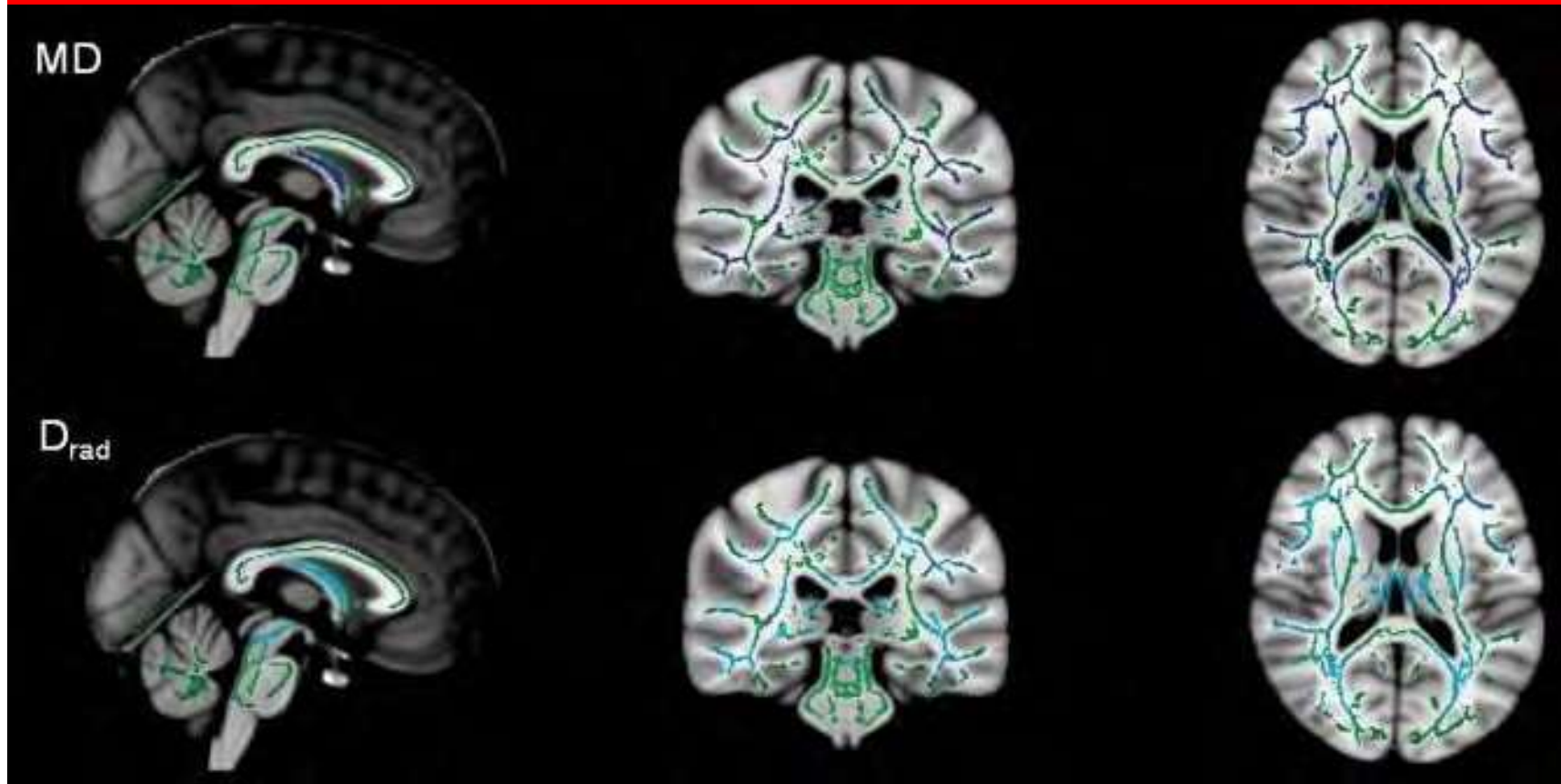
“Hidden in plain sight”

What’s next ?

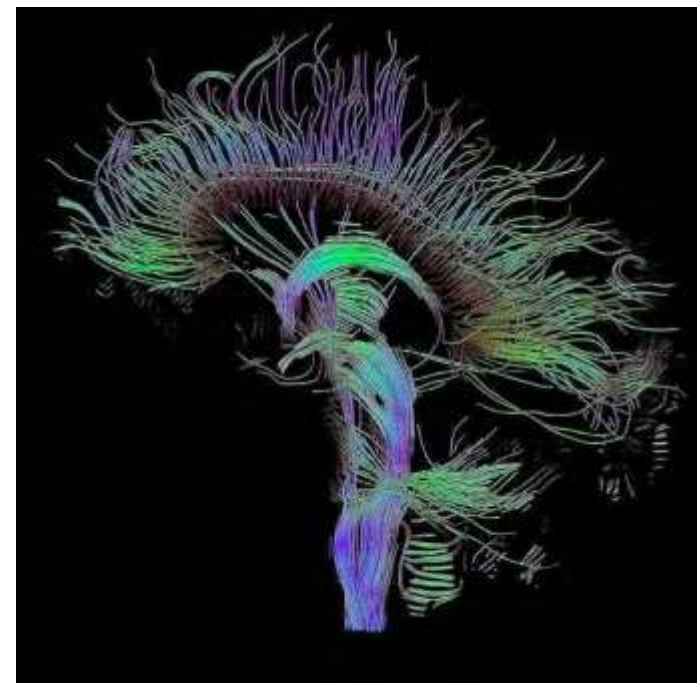
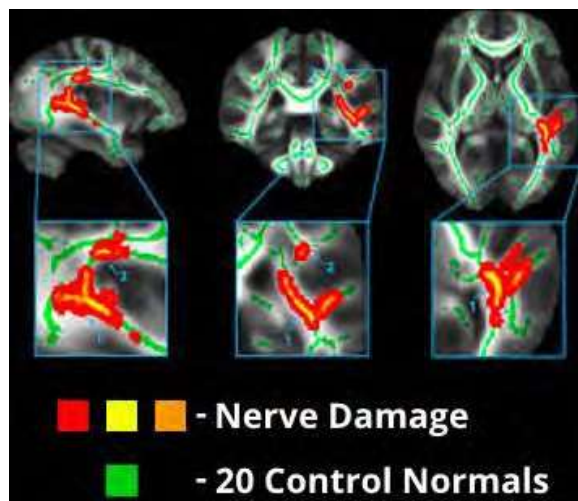


Kinnunen KM et al. (2011)

White matter damage and cognitive impairment after traumatic brain injury. *Brain* 134:449-463.



Diffusion Tensor Imaging DTI



- Objective method to detect nerve damage in brain caused by mTBI
- Measures directional movement of water molecules moving along nerve pathways
- Disruption of H₂O pathway => injury to nerve pathway = **microscopic** white matter tract injury

WM Tractography or CONNECTIVITY or “which bit of brain connected to which other part”

Should lawyers be using DTI in evidence ?

- Detects evidence of DAI but sensitivity limited/ misleading
- Imaging must be integrated with complete clinical picture.
- Future recovery or decline cannot be predicted from DTI
- Hard to standardise
- Not specific - exclusions include:
 - Neurological & psychiatric diseases*
 - Drugs*
 - Previous TBI*
 - Systemic illness.*

Diffusion Kurtosis Imaging

- Poor specificity of DTI limits clinical application
- DKI evaluates more complex 3D diffusion pattern
- More sensitive tool to identify brain injury
- May prove helpful in predicting cognitive outcome
- But 7-10 minute sequences

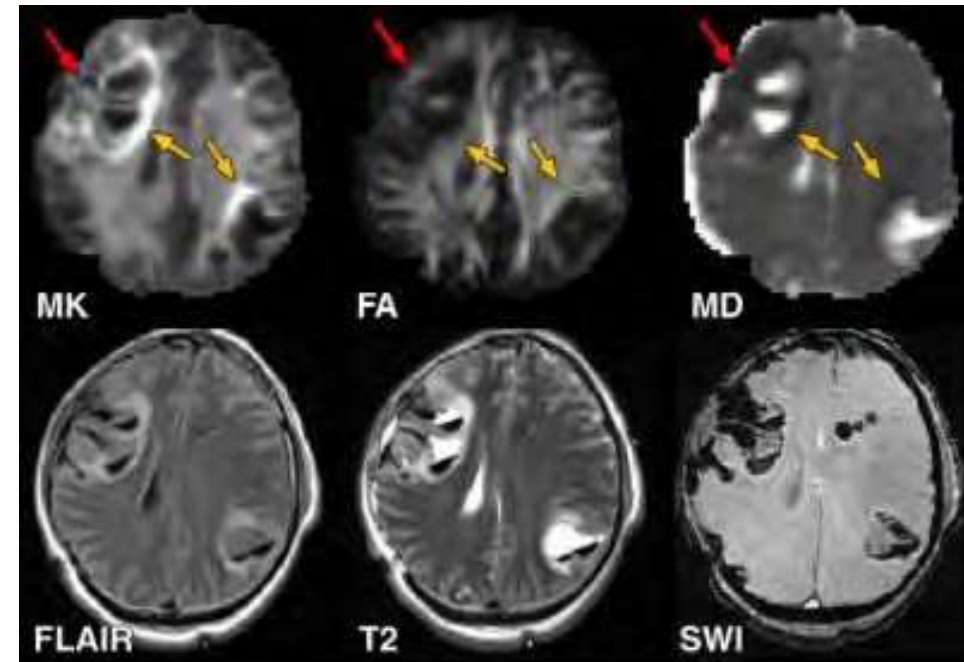
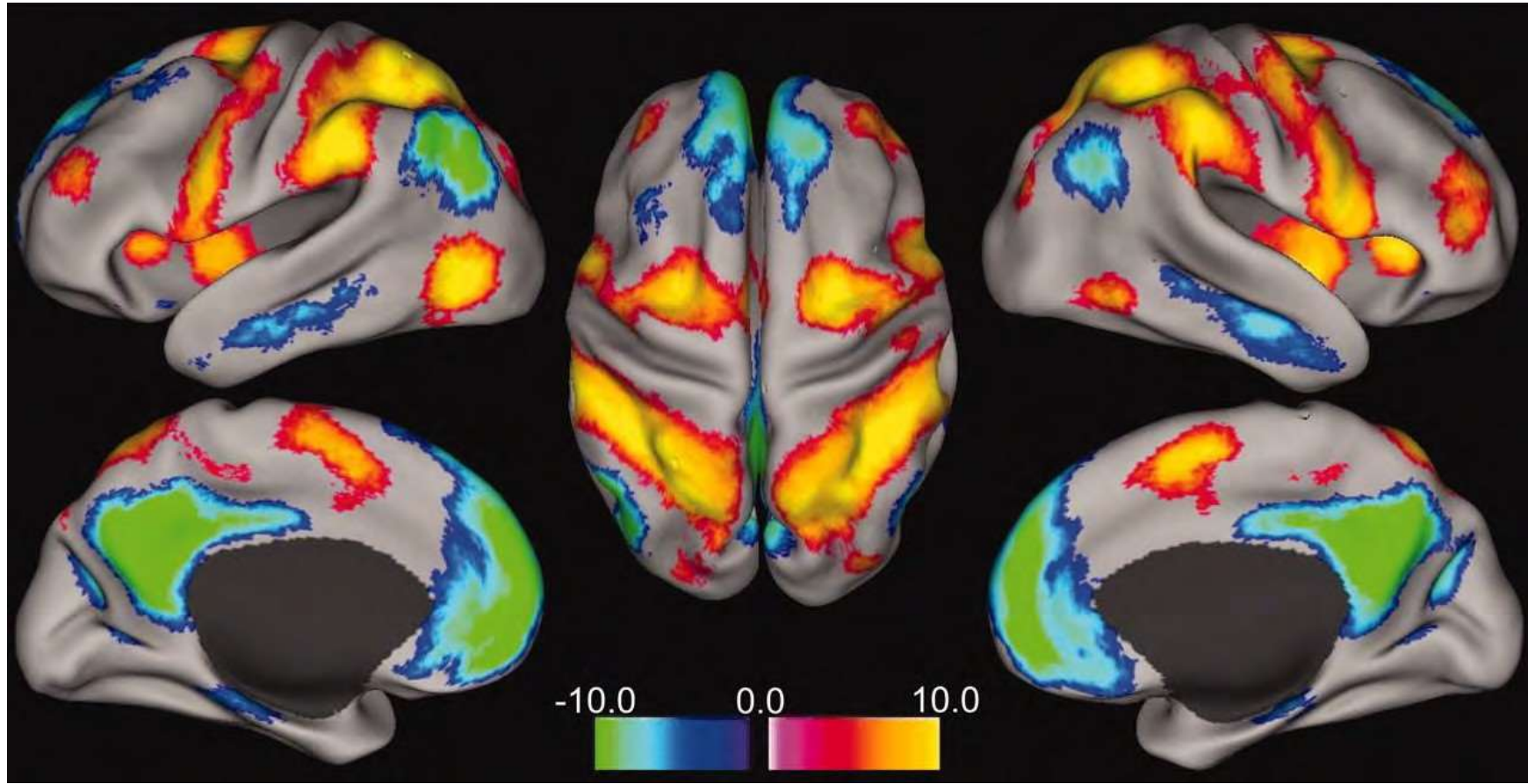
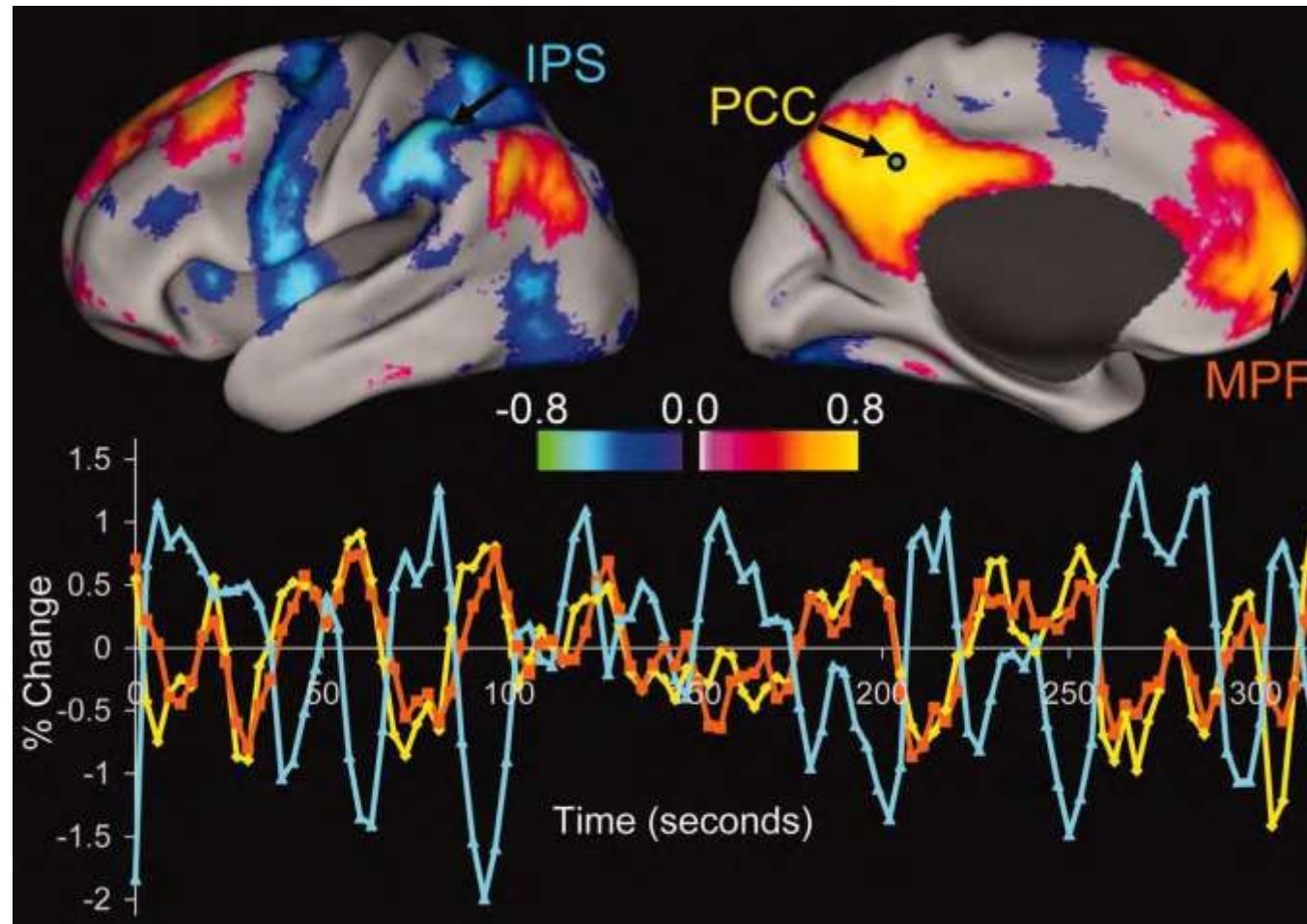


Fig. 5 —40-year-old man with traumatic brain injury sustained in fall from roof (Glasgow coma score, 3t) who underwent MRI 2 days after injury. Diffusion kurtosis parameter maps (MK = mean kurtosis, FA = fractional anisotropy, MD = mean diffusion) and conventional MR images (FLAIR, T2-weighted, SWI = susceptibility weighted) at same axial location. Extremely high degrees of MK may indicate formation of glial scars around lesions (*yellow arrows*), which are otherwise not noticeable on conventional images or FA map. High MK regions are accompanied by restricted diffusion as indicated on MD map, but different degrees of contrast (*red arrows*) on MD and MK maps are evident.

fMRI

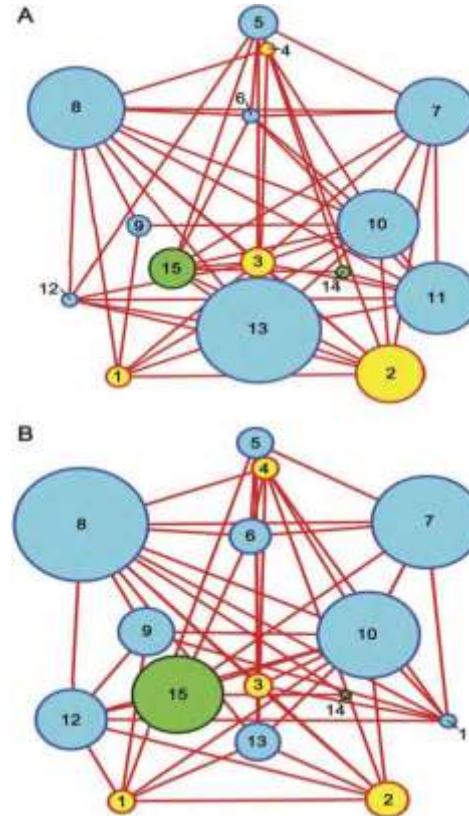


fMRI



Traumatic brain injury impairs small-world topology.

Pandit AS et al. (2013) Neurology 80:1826-1833

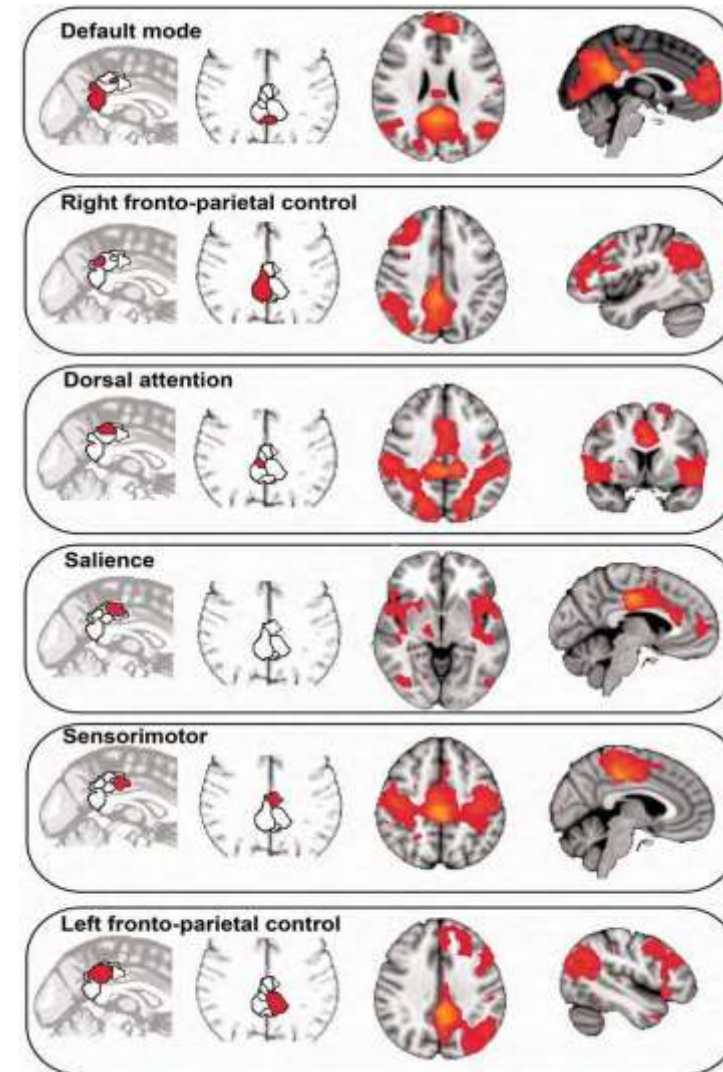
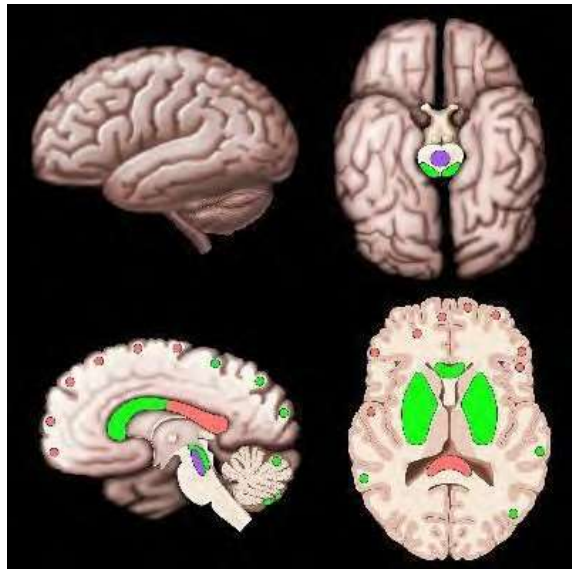
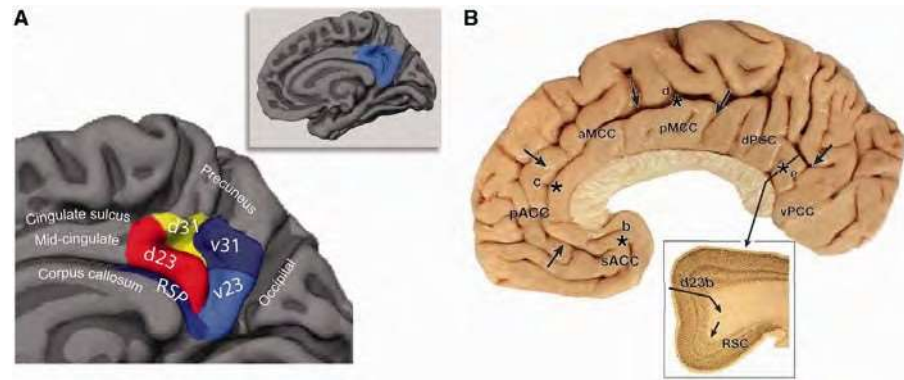


Topographic connectivity maps in normal control and traumatic brain injury patient groups.

Network connectivity in (A) normal controls and (B) traumatic brain injury patients.

The role of the posterior cingulate cortex in cognition and disease.

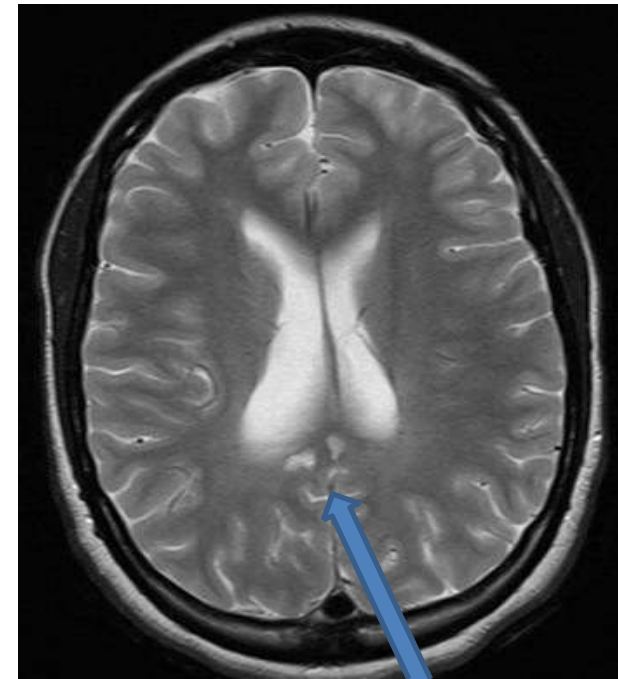
Leech R and Sharp DJ (2014) Brain 137:12-32.



Teenager knocked down by car travelling 70mph

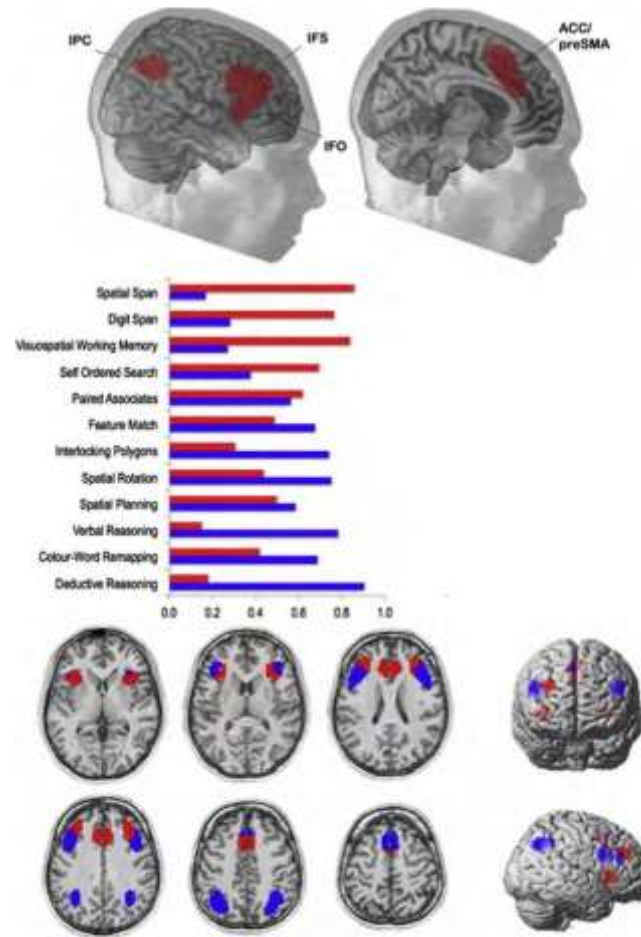
Marked cognitive and behavioural sequelae

CT & MRI = “tip of the iceberg”



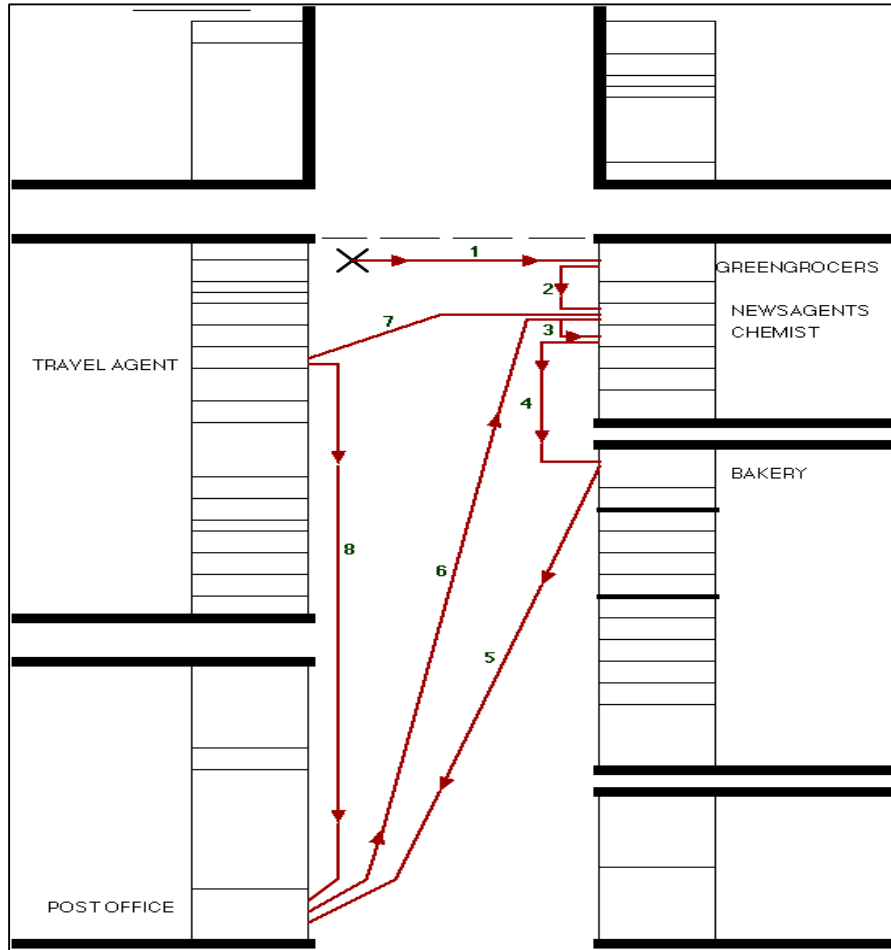
Fractionating human intelligence.

Neuron Hampshire A et al. (2012) 76:1225-1237.

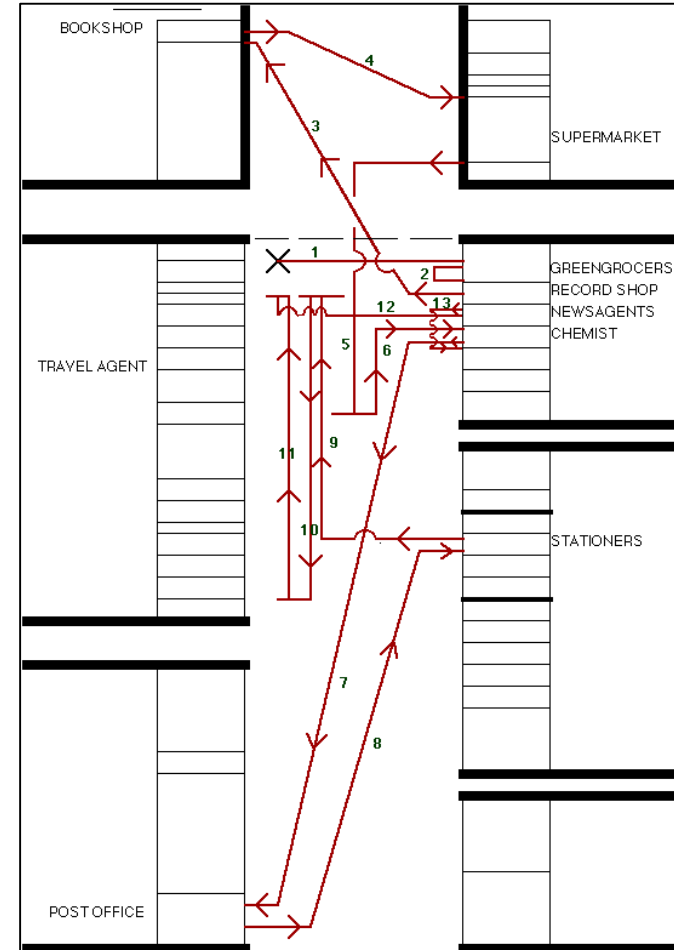


Multi-tasking

- Tasks
 - Buy 6 items (e.g. lettuce, a bar of soap)
 - Find out 4 pieces of information (e.g. coldest place in Britain yesterday)
 - Be outside travel agent shop in 15 minutes
 - Rules
 - Spend as little money and take as little time as possible
 - Don't enter a shop except to buy something
 - Don't use anything not bought on the street (except a watch)
 - Inform experimenter of purchases straight after leaving a shop
- Multiple Errands Test**



Healthy volunteer



TBI patient

Examples of errors made only by patient

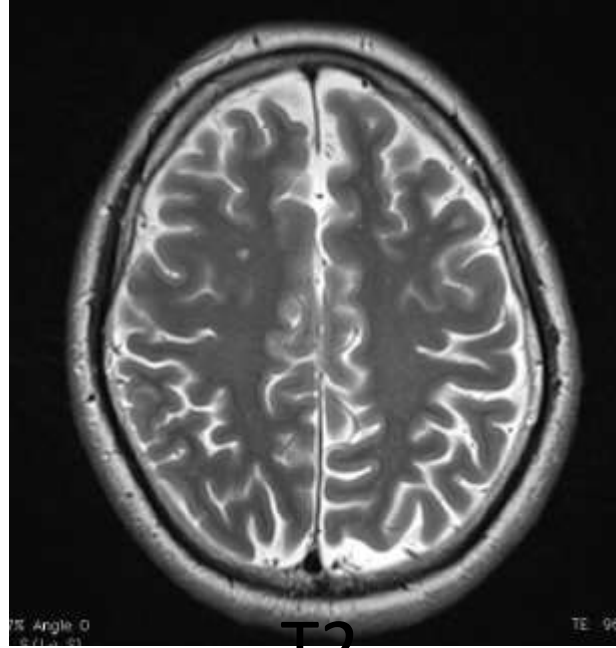
- Recorded incorrect information
- Ate purchase (chocolate) before test finished
- Left boundaries of shopping centre
- Jumped queue at payment point
- Swore loudly
- Removed shoes
- Purchased pornographic magazine

How does this information influence medicolegal cases today ?

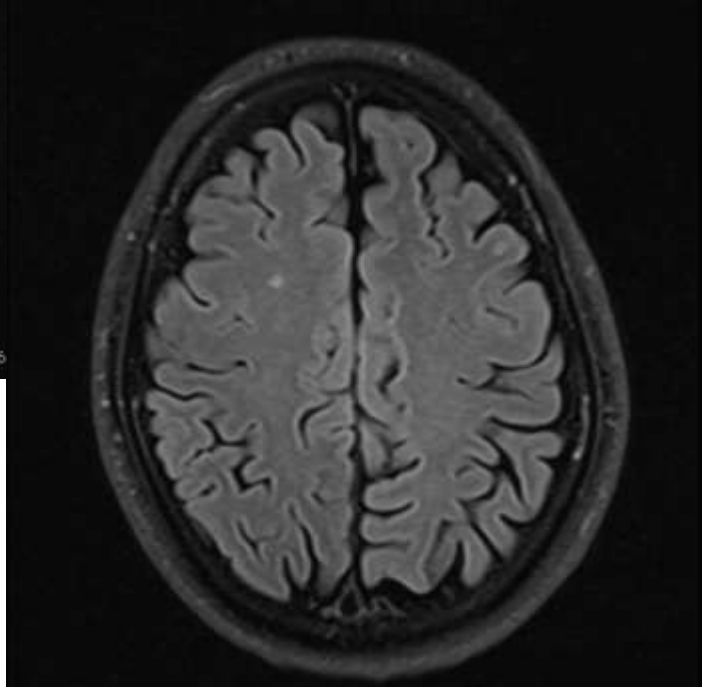
45 years company director

Knocked off motor scooter at 40 m.p.h., wearing helmet.

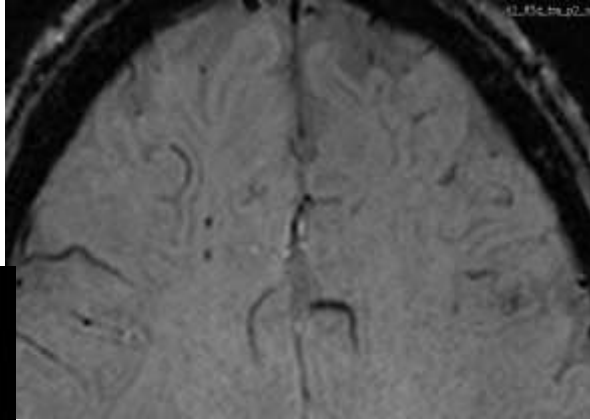
- Ruptured spleen.
- Lacerated liver.
- Eight fractured ribs.
- LOC 10 minutes, 5 minutes RGA, 30 minutes PTA.
- Neuropsychology 6 months later – impairment of executive function.
 - inhibition
 - Flexibility
 - Planning
 - Multi-tasking
 - Judgement
 - Mood disturbance



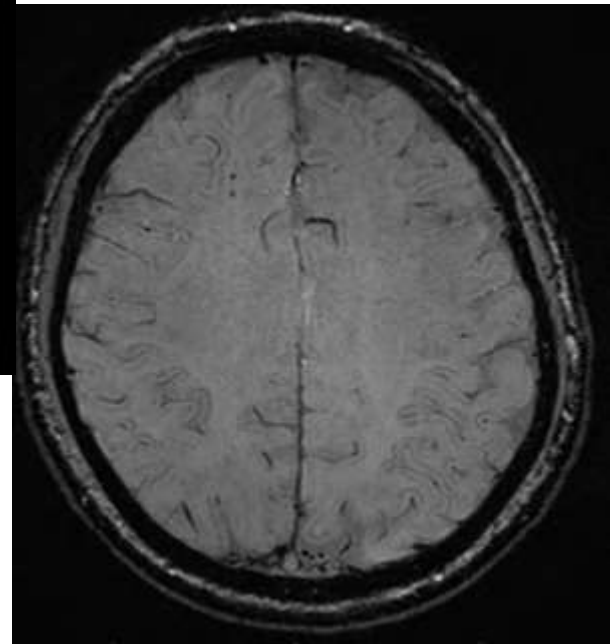
T2



FLAIR



3T SWI



Confirmed DAI / White matter tract disruption:

Settled for £2.6million.

Chronic traumatic encephalopathy CTE

“Hidden in plain sight”

Risk factors:

Multiple mTBI + genetic, stress, drink/drugs, other.

Multiple Acute mTBI:

DAI / microvascular disruption / neuroinflammation

Deposition of small amounts of tau protein

CTE = focal deposition of tau that spreads in the cortex => medial temporal lobe => brainstem in susceptible individuals.

The mechanisms underlying the spread of tau and the accumulation of TDP-43 in the brain remains to be defined.

NFL exec admits to CTE-football head trauma link

Mar. 15, 2016 10:58 AM EDT

NFL Study: Players Face Huge Risk of Dementia

*Former NFL players 30–49: memory-related diseases 19x national population.
6.1% players >50 diagnosed with dementia, 5x Nat average 1.2%.*

WASHINGTON (AP) — An NFL official has acknowledged a link between football and the brain disease CTE for the first time.

Jeff Miller, the NFL's senior vice president for health and safety:

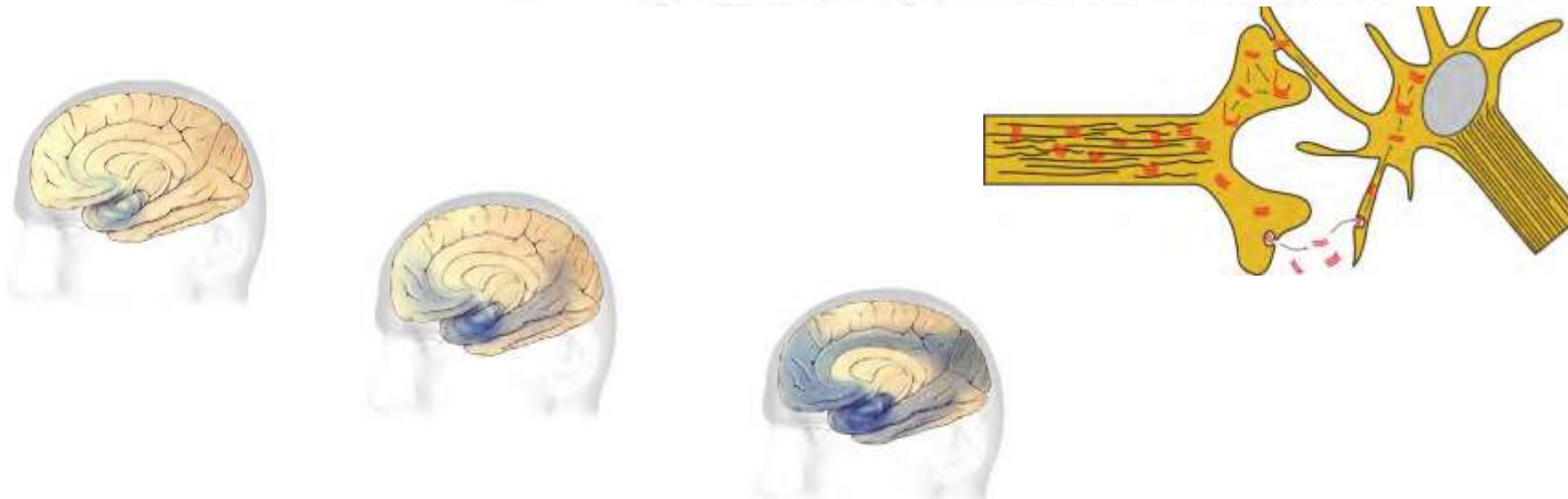
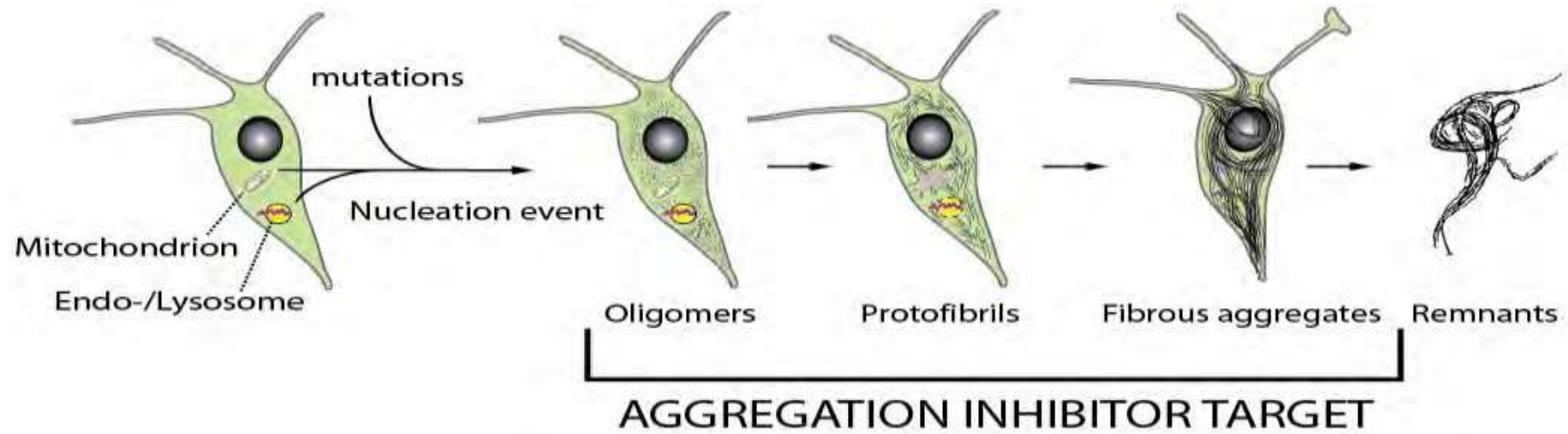
"Do you think there is a link between football and degenerative brain disorders like CTE?"

"Well, certainly, Dr. McKee's research shows that a number of retired NFL players were diagnosed with CTE, so the answer to that question is certainly 'yes,'...."

NFL acknowledge CTE as a disease linked to repeated brain trauma and associated memory loss, depression and progressive dementia.

<http://bigstory.ap.org/urn:publicid:ap.org:7fd994cea9734899a27ef57ffc45c7e6>

Spread of pathology and Braak staging



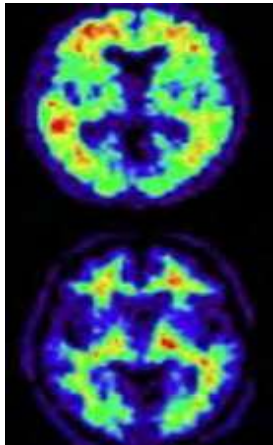
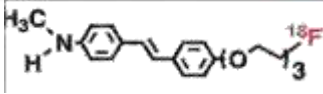
Can we detect CTE ?

PET MRI Specific tau and amyloid binding agents

β -amyloid: Approved tracers by CHMP/FD

Tau: awaited in research only

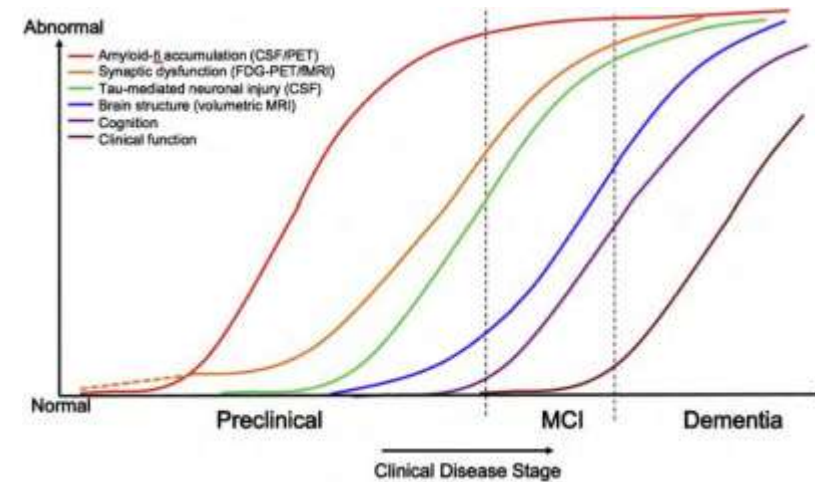
Florbetapir 18F



Developed for AD

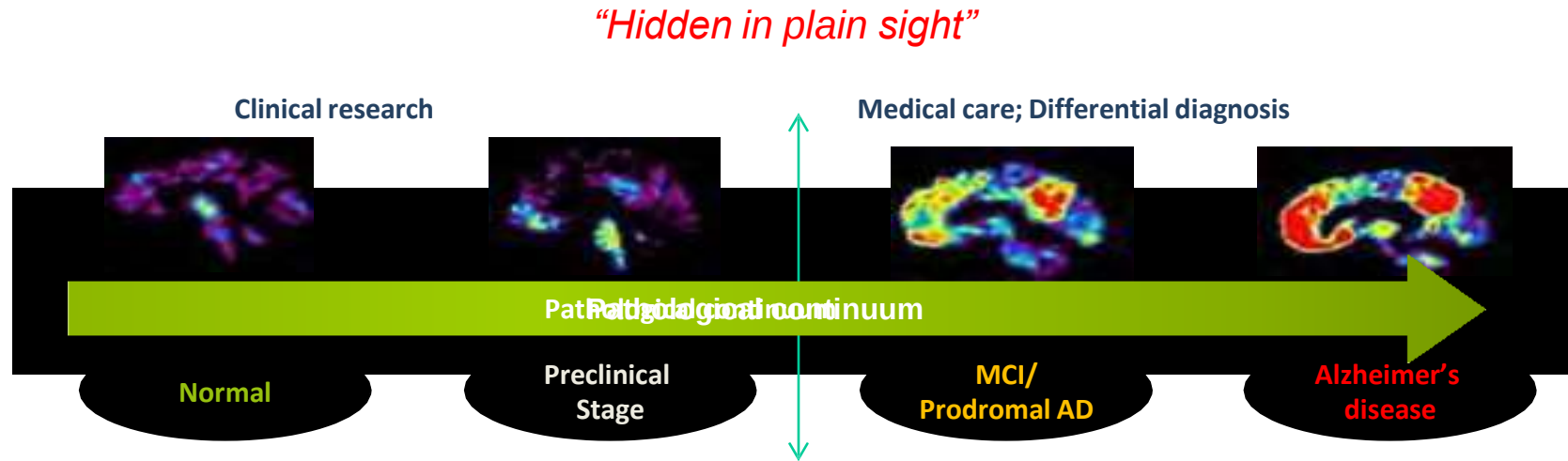
Cognitively
Normal
Subjects

Repeated TBI in young sport's men



Florbetaben and Florbetapir are approved for clinical read in grey scale only. Flutemetamol is approved for clinical read in color only.

Overcoming the clinical challenges new paradigm CTE cf AD ?



Like AD, CTE is a clinical-biological entity with a clinical phenotype ranging from normal cognition to severe dementia:

- Specific symptoms imply that we are in the clinical spectrum of the continuum
- Biomarkers, if correlate with pathology, indicate that we are within the continuum

Brain imaging in summary 2016

How to find brain injury

What does it imply for the future ?

TBI

CT:

Acute macroscopic structural brain damage

3T MRI:

Structural brain damage

Connectivity damage- tractography

Quantitative pathophysiological damage

DWI, SWI DTI, DKI fMRI & MEG

Newer techniques & correlate with neurocognitive outcome in mTBI

Multiple TBI / CTE

PET CT / MRI

Specific tau binding agents

Amyloid binding agents

Indicator of risk for progressive neurodegenerative disease in the future

IMPLICATIONS of Moving to evidence based quantitative practice

**CTA/MRA/ Angiography others not discussed*

Re:Cognition Health

COMPLETE COGNITIVE CARE

Thank you

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Acknowledgement: Illustrations on slides 11,12,14,17 are from Anne Osborn e-textbook Neuroradiology Differential Diagnosis and Neuroradiology.